

☁️ TOP 50 GCP CONCEPT QUESTIONS ☁️

1 What is Google Cloud Platform (GCP)?

GCP is a suite of cloud computing services offered by Google. It includes computing power, storage, databases, networking, AI/ML, and analytics tools delivered over Google's global infrastructure.

✓ *Key Benefit: Scalable, secure, reliable and pay-as-you-go cloud services.*

2 What are the main services provided by GCP?



Compute Engine



Cloud Storage



BigQuery



Kubernetes Engine



Cloud Networking



AI/ML (Vertex AI)



Think: Compute, Storage, Database, Networking, Big Data, AI/ML, DevOps

3

What is the difference between GCP, AWS, and Azure?

Aspect	GCP	AWS	Azure
Company	Google	Amazon	Microsoft
Strength	Data Analytics, AI/ML, BigQuery	Wide range of services, mature ecosystem	Hybrid cloud, enterprise integration
Pricing	Per-second billing in many services	Pay-as-you-go, Spot instances	Pay-as-you-go, Enterprise agreements
Best For	Data, ML, Startups	Large scale, Global applications	Enterprises, .NET, Hybrid

4

What is a Project in GCP?

A Project is a logical container that holds all your resources and services in GCP. It helps you organize, manage billing, set permissions, and isolate resources.



Remember: All resources in GCP must belong to a Project.

5

What is an Organization in GCP?

An Organization is the top-level entity that represents your company. It allows you to manage multiple projects, users, billing accounts, and policies centrally.



Key Point: Organization → Folder → Project → Resources



Golden Rule: Organization is for management, Project is for usage!

 COMPUTE SERVICES**Compute Engine**

Virtual machines (VMs) running on Google's infrastructure. Full control over OS, applications, and environment.

✓ *Use Case: Web hosting, database servers, high-performance computing, legacy app migration*

🔧 *Basic Setup: Create VM instance → Choose machine type → Select OS → Configure firewall rules → SSH & deploy*

Pricing: Per-second billing, discounts for sustained use

**Google Kubernetes Engine (GKE)**

Managed Kubernetes service for containerized applications. Auto-scaling, load balancing, and self-healing.

✓ *Use Case: Microservices, containerized apps, CI/CD pipelines, scalable workloads*

🔧 *Basic Setup: Create cluster → Push Docker image to Container Registry → Deploy pods → Configure services*

Pricing: Pay for nodes + minimal cluster fee

**Cloud Run**

Serverless container platform. Run stateless containers without managing servers.

✓ *Use Case: REST APIs, event-driven functions, background jobs, microservices*

🔧 *Basic Setup: Containerize app → Push to Container Registry → Deploy Cloud Run service → Get auto URL*

Pricing: Per request pricing, free tier included

**STORAGE & DATABASE SERVICES**



Cloud Storage

Scalable object storage for unstructured data (files, images, videos, backups).

✓ *Use Case:* Data lakes, media files, backups, static website hosting, data archives

🔧 *Basic Setup:* Create bucket → Set storage class → Configure access control → Upload objects

Pricing: Storage + egress charges, Coldline/Archive for cheaper archival



Cloud SQL

Fully managed relational database (MySQL, PostgreSQL, SQL Server). Automatic backups and high availability.

✓ *Use Case:* Web apps, transactional data, OLTP systems, traditional databases

🔧 *Basic Setup:* Create instance → Choose DB type/version → Set machine type → Create database → Connect from app

Pricing: Per instance + storage + backup charges



Firestore & Cloud Datastore

NoSQL document databases. Firestore is the newer, more scalable version with real-time sync.

✓ *Use Case:* Mobile apps, real-time collaboration, user profiles, content management, IoT

🔧 *Basic Setup:* Enable Firestore → Create database → Set security rules → Start writing documents

Pricing: Per read/write/delete operations + storage

 DATA & ANALYTICS SERVICES **BigQuery**

Serverless data warehouse for analyzing massive datasets. Lightning-fast SQL queries on petabytes of data.

✓ *Use Case:* Data analytics, business intelligence, log analysis, machine learning datasets

🔧 *Basic Setup:* Create dataset → Load data (CSV/JSON/Parquet) → Write SQL queries → Visualize results

Pricing: Per GB queried (first 1 TB free/month), flat-rate pricing available

 Cloud Dataflow

Managed batch and stream data processing service. Unified API for data pipelines (Apache Beam).

✓ *Use Case:* ETL pipelines, real-time data processing, data transformation, log processing

🔧 *Basic Setup:* Write Beam pipeline → Submit job → Monitor execution → Store results in BigQuery/Storage

Pricing: Per vCPU-hour + storage charges

 Cloud Pub/Sub

Messaging service for event streaming. Decouple services with pub/sub architecture.

✓ *Use Case:* Real-time messaging, IoT events, log processing, microservices communication

🔧 *Basic Setup:* Create topic → Create subscription → Publish messages → Subscribe and consume

Pricing: Per GB published/pulled

 Cloud Dataproc

Managed Hadoop/Spark clusters. Run distributed data processing on Spark, Hadoop, Hive, Pig.

✓ *Use Case: Big data processing, Spark jobs, legacy Hadoop migrations, batch jobs*

🔧 *Basic Setup: Create cluster → Submit Spark job → Monitor execution → Auto-scaling applies*

Pricing: Per node-hour + disk charges



AI/ML SERVICES



Vertex AI

Unified ML platform for building, training, and deploying ML models. Pre-trained models + custom training.

✓ *Use Case: Computer vision, NLP, forecasting, custom ML models, AutoML for beginners*

🔧 *Basic Setup: Prepare dataset → Train model (AutoML or custom) → Evaluate → Deploy to endpoint*

Pricing: Per training hour + per prediction



Pre-trained Models (Vision, Language, etc.)

Use Google's pre-built models for common tasks: image recognition, text analysis, translation, speech-to-text.

✓ *Use Case: Quick AI implementation without ML expertise, document processing, image analysis*

🔧 *Basic Setup: Install SDK → Call Vision/Language API → Send image/text → Get results*

Pricing: Per request or per 1000 requests

 NETWORKING SERVICES

Virtual Private Cloud (VPC)

Private network for GCP resources. Control IP ranges, subnets, routing, and firewall rules.

✓ *Use Case:* Network isolation, multi-tier architectures, hybrid cloud setup, security segmentation

🔧 *Basic Setup:* Create VPC → Define subnets → Set firewall rules → Configure Cloud Router for hybrid

Pricing: Free (minimal charges for advanced features)



Cloud Load Balancing

Distribute traffic across resources. Global load balancing with low latency routing.

✓ *Use Case:* High availability, auto-scaling, geographic distribution, traffic management

🔧 *Basic Setup:* Create backend service → Configure health checks → Create load balancer → Assign target groups

Pricing: Per forwarding rule + data processing charges



Cloud CDN

Content Delivery Network. Cache content globally for faster delivery to users.

✓ *Use Case:* Static content distribution, media streaming, website acceleration, API response caching

🔧 *Basic Setup:* Enable CDN on load balancer → Set cache policies → Configure cache keys

Pricing: Per GB cached/served



Cloud Armor

DDoS protection and web application firewall. Protect against malicious traffic and attacks.

✓ *Use Case:* DDoS defense, WAF rules, geo-blocking, rate limiting, bot protection



Basic Setup: Create security policy → Add rules → Attach to load balancer → Monitor threats

Pricing: Free for basic DDoS, charges for advanced WAF rules



SECURITY & MANAGEMENT



Cloud KMS (Key Management Service)

Manage encryption keys. Hardware security module (HSM) backed key management.



Use Case: Data encryption, compliance requirements, key rotation, secret management



Basic Setup: Create keyring → Create key → Grant permissions → Use in encryption operations

Pricing: Per key version + per encryption/decryption operation



Identity & Access Management (IAM)

Control who has access to what. Role-based access control (RBAC) for GCP resources.



Use Case: Permission management, least privilege access, audit trails, resource sharing



Basic Setup: Create service accounts → Assign roles → Create key pairs → Use in applications

Pricing: Free (included with GCP)



Cloud Audit Logs

Track activity and changes to GCP resources. Compliance and security monitoring.



Use Case: Compliance auditing, security investigation, change tracking, forensics



Basic Setup: Enable audit logging → Export logs to BigQuery/Storage → Create dashboards

Pricing: Storage charges for logs, free first 50GB/month



DEPLOYMENT & MANAGEMENT SERVICES



Container Registry & Artifact Registry

Store and manage Docker container images. Vulnerability scanning and access control.

✓ *Use Case:* Docker image storage, CI/CD pipelines, container deployment, multi-region distribution

🔧 *Basic Setup:* Configure Docker authentication → Push image → Pull in deployment → Deploy to GKE/Run

Pricing: Storage charges + network egress



Cloud Build

Serverless CI/CD platform. Build, test, and deploy applications automatically.

✓ *Use Case:* Continuous integration, automated testing, Docker builds, deployment automation

🔧 *Basic Setup:* Create cloudbuild.yaml → Connect repo → Push commit → Automatic build & deploy

Pricing: Free 120 minutes/day, then per minute charges



Cloud Functions

Serverless functions for event-driven computing. Run code without managing servers.

✓ *Use Case:* HTTP APIs, event handlers, background jobs, webhooks, scheduled tasks

🔧 *Basic Setup:* Write function → Deploy with CLI → Configure trigger → Function auto-scales

Pricing: Per million invocations + compute time (free tier available)



Cloud Monitoring & Logging

Monitor application performance and system health. Centralized logging and alerting.

✓ *Use Case:* Application monitoring, performance tracking, alerting, troubleshooting, SLO tracking



Basic Setup: Enable monitoring → Create dashboards → Set alert policies → Route notifications

Pricing: Free first 50GB logs/month, storage charges after



DEVELOPER & INTEGRATION SERVICES



API Management

Create, manage, and secure APIs. Rate limiting, authentication, versioning, analytics.

✓ *Use Case: Public APIs, microservice gateways, third-party integrations, API monetization*



Basic Setup: Create API → Define methods → Set policies → Deploy proxy → Monitor usage

Pricing: Subscription based + per policy calls



Cloud Tasks

Manage large-scale task processing. Guaranteed delivery and retry logic.

✓ *Use Case: Job queuing, distributed task execution, asynchronous processing, deferred jobs*



Basic Setup: Create queue → Create tasks → Cloud Tasks handles retry & delivery

Pricing: Per million task operations



GCP Quick Reference Checklist

- ▶ *Getting Started: Create GCP Project → Set up billing → Enable APIs → Configure IAM*
- ▶ *Common Architecture: VPC → Load Balancer → Cloud Run/GKE → Cloud SQL → Cloud Storage*
- ▶ *For Web Apps: Cloud Run or GKE + Cloud SQL + Cloud Storage + Cloud CDN*
- ▶ *For Data Analytics: Cloud Storage → Dataflow → BigQuery → Data Studio visualization*
- ▶ *For ML Projects: BigQuery (data) → Vertex AI (training) → Cloud Endpoints (deploy)*
- ▶ *Security First: Use VPC, IAM, Cloud KMS, Cloud Armor, enable Audit Logs*

► *Cost Optimization: Use sustained use discounts, set budgets, monitor in Cloud Monitoring*